********** QUESTIONS and SOLUTIONS **********

Information for Candidates

Numerical answers

- Numerical answers should be given as fully evaluated decimal values and not as unevaluated arithmetic expressions.

Notation

The following notation is used in this paper:

- The voltage waveform at node X in an AC circuit is denoted by $x(t)$ while the phasor voltage is denoted by X.

- Component and source values are generally given in Ohms, Farads, Henrys, Volts or Amps with the unit symbol omitted. Where an imaginary number is specified, it represents the complex impedance or phasor value.

- Times are in seconds unless otherwise stated.

Answers

The Questions

1. a) Using nodal analysis, calculate the voltages at nodes *X* and *Y* of Figure 1.1. [6]

KCL at X gives: $\frac{X-15}{6}+\frac{X}{3}+\frac{X-Y}{2}=0 \Rightarrow 6X-3Y-15=0.$

KCL at Y gives: $\frac{Y-X}{2}+\frac{Y+10}{1}-5=0 \quad X-3Y-10=0.$

Solving the simultaneous equations on the right gives X = 1, Y = –3. *[6]*

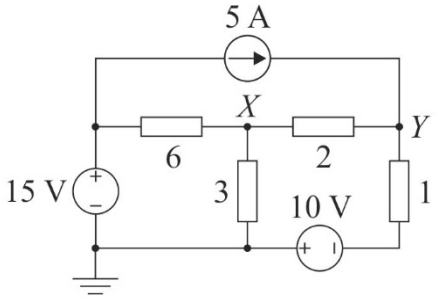


Figure 1.1

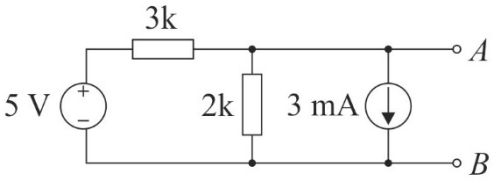


Figure 1.2

b) Draw the Thévenin equivalent of the network shown in Figure 1.2, taking care to show clearly the sign of the voltage source. [5]

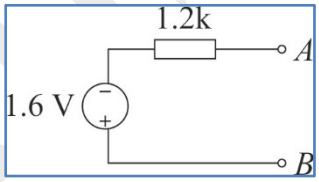
We will make B the reference terminal and find the open-circuit voltage and the short-circuit current. To get V_{OC} *we do KCL at A with the output open-circuit which gives:*

$$\frac{V_{OC}-5}{3}+\frac{V_{OC}}{2}+3=0 \Rightarrow 5V_{OC}+8=0 \Rightarrow V_{OC}=-8/5=-1.6\ V.$$

To get I_{SC} *(arrow from B to A) we do KCL at A with the output shorted which gives:*

$$-I_{SC}+3-\frac{5}{3}=0 \Rightarrow I_{SC}=4/3\ mA.$$

The Thévenin resistance is $R_{Th}=-V_{OC}/I_{SC}=6/5=1.2\ k\Omega$*, so the Thévenin equivalent is:*



[5]

c) For the circuit in Figure 1.3, write down an expression relating *Z* to *X* and *Y*. You should assume the op-amp is ideal. [5]

Using the potential divider rule and superposition, the voltages at the op-amp inputs are:

$$V+=\frac{Y}{2} \quad and \quad V-=\frac{3X}{4}+\frac{Z}{4}.$$

Since there is negative feedback, we also have $V+=V-$*, so:*

$$\frac{Y}{2}=\frac{3X}{4}+\frac{Z}{4} \Rightarrow Z=-3X+2Y.$$ *[5]*

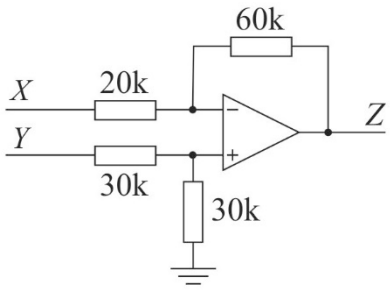


Figure 1.3

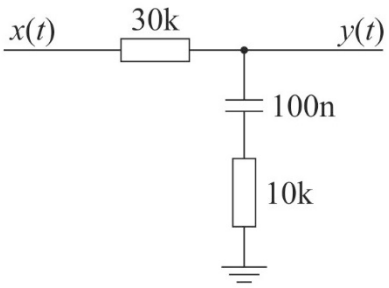


Figure 1.4

d) Write down the transfer function $\frac{Y}{X}(j\omega)$ for the circuit in Figure 1.4. Using the transfer function, or otherwise, derive an expression for the response $y(t)$ for $t > 0$ when $x(t)$ has the form:

$$x(t) = \begin{cases} 0 & t < 0 \\ 2 & t \geq 0 \end{cases}.$$ [8]

The transfer function is: $H = \frac{Y}{X} = \frac{R+\frac{1}{j\omega C}}{R+\frac{1}{j\omega C}+3R} = \frac{1+j\omega\tau}{1+j4\omega\tau}$

where $\tau = 10k \times 100n = 1$ *ms.* *[3]*

From the expression for H we can see that the low- and high-frequency gains are 1 and 0.25 respectively, and the time constant in the transient response will be $4\tau = 4$ *ms.*

The input voltage is DC either side of the step at t = 0, so the steady state responses are $y_{ss}(t) = H_{LF} \times 0 = 0$ *for* $t < 0$ *and* $y_{ss}(t) = H_{LF} \times 2 = 2$ *for* $t > 0$*. Also, the input step is* $\Delta x = +2$*, so the output step is* $\Delta y = H_{HF} \times 2 = 0.5$*. We therefore have* $y(0+) = 0.5$*, and the transient amplitude A must be such that* $2 + A = 0.5$*.*

The solution for t > 0 is therefore:

$$y(t) = 2 - 1.5e^{-250t}.$$ *[5]*

Question 1 continues on the next page…

Answers

Question 1 continued

e) For the circuit in Figure 1.5, choose the values of R_E and R_C to give a collector bias current of 0.4 mA and a collector voltage of 6 V. State clearly any assumptions made in your calculations. [5]

It is convenient to transform the bias network into a Thévenin source which gives $V_{Th} = 20\times10/130 = 1.538$ V and $R_{Th} = 20\times110/130 = 16.923$k. KVL in the loop containing the bias source, the EBJ and R_E then gives:

$$I_E R_E + V_{BE} + \frac{I_E}{1+\beta} R_{Th} = V_{Th} \quad \Rightarrow \quad R_E = \frac{V_{Th} - V_{BE}}{I_E} - \frac{R_{Th}}{1+\beta}$$

We have assumed here that the transistor is active (as it is bound to be from values of V_{Th} and V_C). We require an emitter current of $I_E = I_C/\alpha = 0.4\times201/200 = 0.402$ mA, and if we assume $V_{BE} = 0.7$ V the above equation gives $R_E = 2.0$k. The collector voltage is $V_C = 10 - R_C I_C$, so for $V_C = 6$ V we require $R_C = 10$k. *[5]*

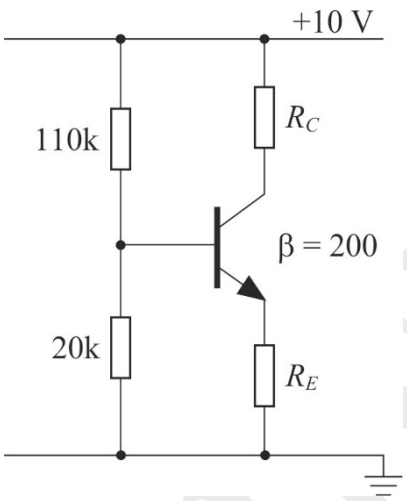


Figure 1.5

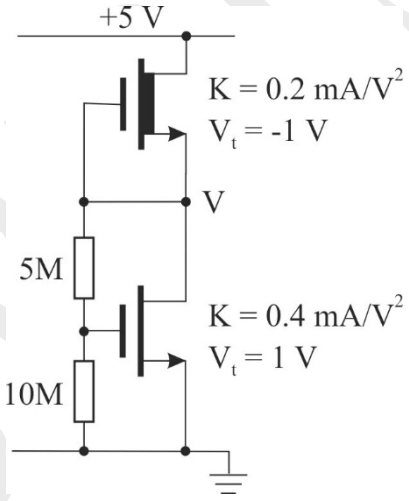


Figure 1.6

f) Calculate the voltage V in the circuit of Figure 1.6, assuming the MOSFETs have infinite output resistance. State any other assumptions made in your calculations. [5]

The bias current is set by the upper FET which we will refer to as Q2. This device has $V_{GS} = 0$, so assuming it is active we have $I_{D2} = K_2(-V_{t2})^2 = 0.2$ mA. The current in the resistor network is tiny in comparison so we will neglect it. The gate voltage of Q1 will then be that required to give a drain current of 0.2 mA, i.e.:

$$V_{GS1} = V_{t1} + \sqrt{0.2m/K_1} = 1 + \frac{1}{\sqrt{2}} V$$

The voltage V is then obtained as $V = 1.5 \times V_{GS1} = 2.56$ V.
With this value of V, the $V_{DS2} = 2.44 > -V_{t2}$, so our assumption about the mode of Q2 was correct. *[5]*

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g) Using the resistance reflection rule, or otherwise, determine the small-signal output resistance of the circuit in Figure 1.7. [6]

Using the resistance reflection rule: $R_o = \frac{R_S}{1+\beta} + r_e = \frac{10k}{201} + \frac{25m}{5m} = 54.75\ \Omega.$ *[6]*

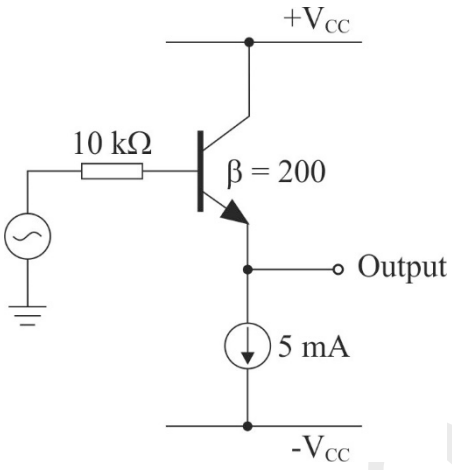


Figure 1.7

Answers

2. In the circuits shown below, U_1 and U_2 are single-supply-rail op-amps with rail-to-rail input and output voltage ranges. You may assume the op-amps are ideal apart from having limited output voltage swing and input voltage range.

a) For the circuit in Figure 2.1, show that the voltages at *X* and *Y* satisfy the relation:

$$\frac{dy}{dt} = \frac{1}{2}\frac{dx}{dt} - \frac{x}{2R_1C}$$

[8]

The + input of the op-amp is at $V^+ = x/2$, and there is negative feedback so we also have $V^- = x/2$. The current in R1 is therefore x/2R1, and from the capacitor equation $i = C\frac{dv}{dt}$ we have: $\frac{x}{2R1} = C\frac{d}{dt}(V^- - y) = C\left(\frac{1}{2}\frac{dx}{dt} - \frac{dy}{dt}\right)$.

Rearranging to make $\frac{dy}{dt}$ the subject gives the required result. *[8]*

b) For the circuit in Figure 2.2, show that the output voltage at *Z* is given by:

$$z = \begin{cases} V_{DD} & y < (1-k)V_{DD} \\ 0 \text{ or } V_{DD} & (1-k)V_{DD} < y < k\,V_{DD}, \\ 0 & y > kV_{DD} \end{cases}$$

and derive an expression for the parameter *k* in terms of R_A and R_C. What determines the value taken by *z* when $(1-k)V_{DD} < y < kV_{DD}$? [8]

This op-amp configuration has positive feedback so the output will always be at one of the supply rails. When $z = V_{DD}$, the voltage at the + input will be:

$$V^+ = \frac{R_A}{R_A \| R_C + R_A} V_{DD} = \frac{R_C + R_A}{2R_C + R_A} V_{DD} = kV_{DD} \quad \text{where } k = \frac{R_C + R_A}{2R_C + R_A}.$$

The output will remain high while $y < kV_{DD}$. However, if y rises above kV_{DD} the output will swing low, and the voltage at the + input will change to:

$$V^+ = \frac{R_A \| R_C}{R_A \| R_C + R_A} V_{DD} = \frac{R_C}{2R_C + R_A} V_{DD} = (1-k)V_{DD}.$$

The output will then remain low while $y > (1\text{-}k)V_{DD}$.

The value of z when y is between the switching thresholds is whatever value was arrived at after the most recent output transition. *[8]*

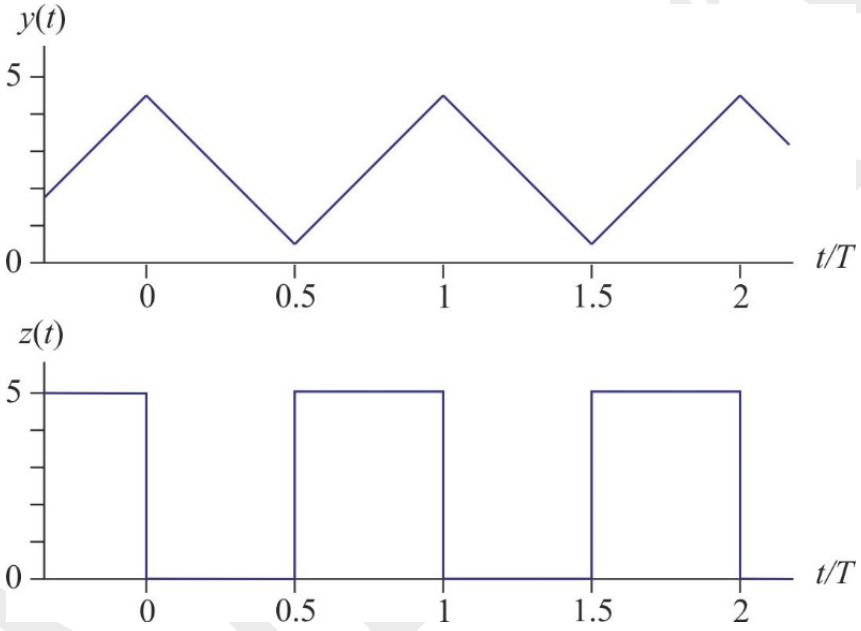
c) The circuits in Figures 2.1 and 2.2 are combined with a MOSFET and one additional resistor to form the circuit shown in Figure 2.3. This circuit is a voltage-controlled oscillator in which the oscillation frequency is proportional to the DC control voltage applied at *X*. You may assume the MOSFET channel has infinite resistance when $z = 0$ and zero resistance when $z = 5$ V.

Answers

Explain how the oscillator works, including in your answer sketches showing the waveforms you would expect at *Y* and *Z* when the control voltage is constant. [8]

When z = 0, U_1 is configured as in part a), and if the control voltage is constant the output y will be a downward linear ramp with a gradient $-\frac{x}{2R_1C}$. The voltage at Y will continue to ramp downwards until the lower comparator threshold is reached. At this point z will switch to V_{DD}, Q_1 will turn on, and the input current to the integrator will change sign (while maintaining the same magnitude). The integrator output y will then ramp upwards with a gradient $+\frac{x}{2R_1C}$ until the upper comparator threshold is reached, at which point z will switch back to zero and the cycle will reppeat. *[5]*

For the circuit in Figure 2.3, k = 112/124 = 0.90, so the comparator thresholds will be 0.5 V and 4.5 V. The waveforms at Y and Z will therefore be as shown below, where T is the oscillation period.



[3]

d) Obtain an expression for the oscillation frequency, and hence choose the value of *C* such that the frequency is 2.5 kHz when *x* = 2.5 V. [6]

The voltage y ramps from one threshold to the other in one half period. So, we have:

$$\frac{x}{2R_1C}\frac{T}{2} = (2k-1)V_{DD} \quad f = \frac{1}{T} = \frac{x}{4R_1C(2k-1)V_{DD}}.$$ *[4]*

Rearranging to make C the subject, and putting f = 2.5 kHz, x = 2.5 V gives C = 6.25 nF. *[2]*

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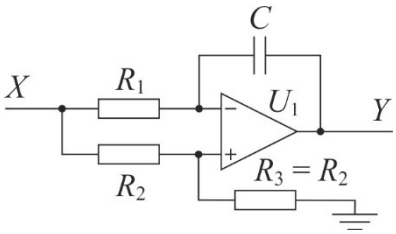


Figure 2.1

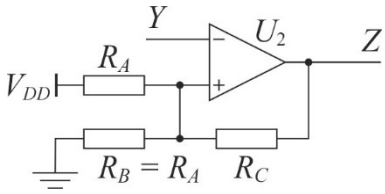


Figure 2.2

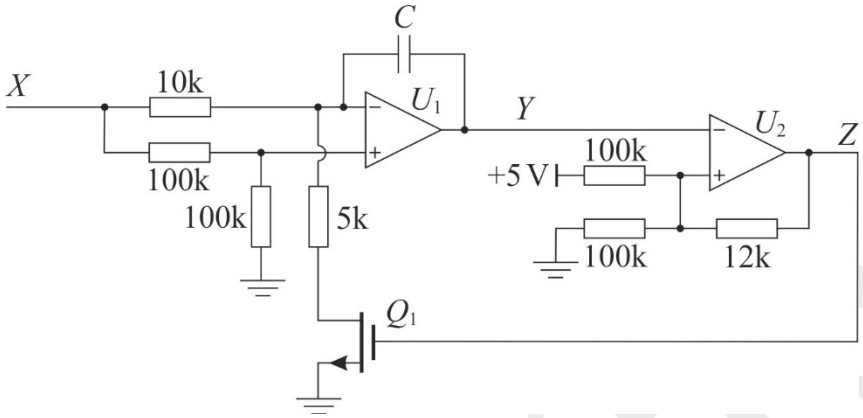


Figure 2.3

Answers

3.

a) Explain (without drawing a diagram) how the circuit of Figure. 3.1 achieves a low common mode gain from the inputs v_{in1} and v_{in2} to the single output terminal, v_{out}.

[4]

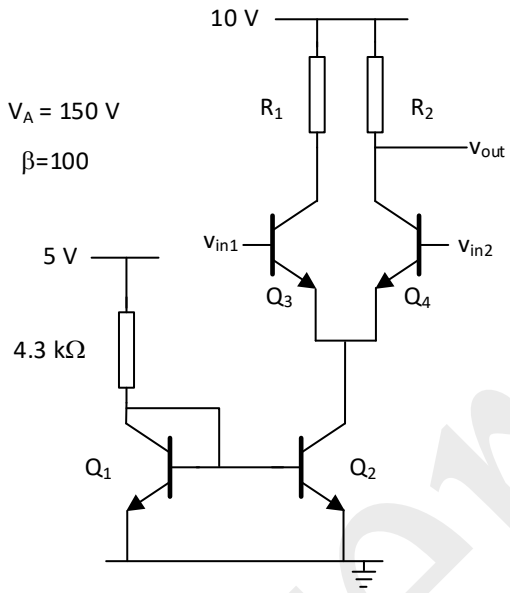


Figure 3.1

Q1 and Q2 form a current mirror, so that the collector current of Q2 is almost constant irrespective of the voltage at the collector of Q2. As long as Vin1 and Vin2 are equal (i.e. there is only a common-mode input voltage being applied), Q3 and Q4 have exactly the same terminal voltages and hence always conduct half the tail current each. Consequently, the voltage drop across R1 and R2 remains constant, irrespective of the common mode input voltage, provided that we stay within the common mode voltage range of the circuit.

[4]

b) Choose a common value for R_1 and R_2 so that the quiescent value of v_{out} is at 5 V.

[2]

The tail current (i.e. the collector current of Q2) is (5-0.7)/4.3k=1mA. Thus, under quiescent conditions, the current in each of Q3 and Q4 is 0.5 mA. We require the output voltage to be at 5 V, so we need to drop 5 V across R1 and R2, which is:
(5 V/0.5 mA)=10 kΩ.

[2]

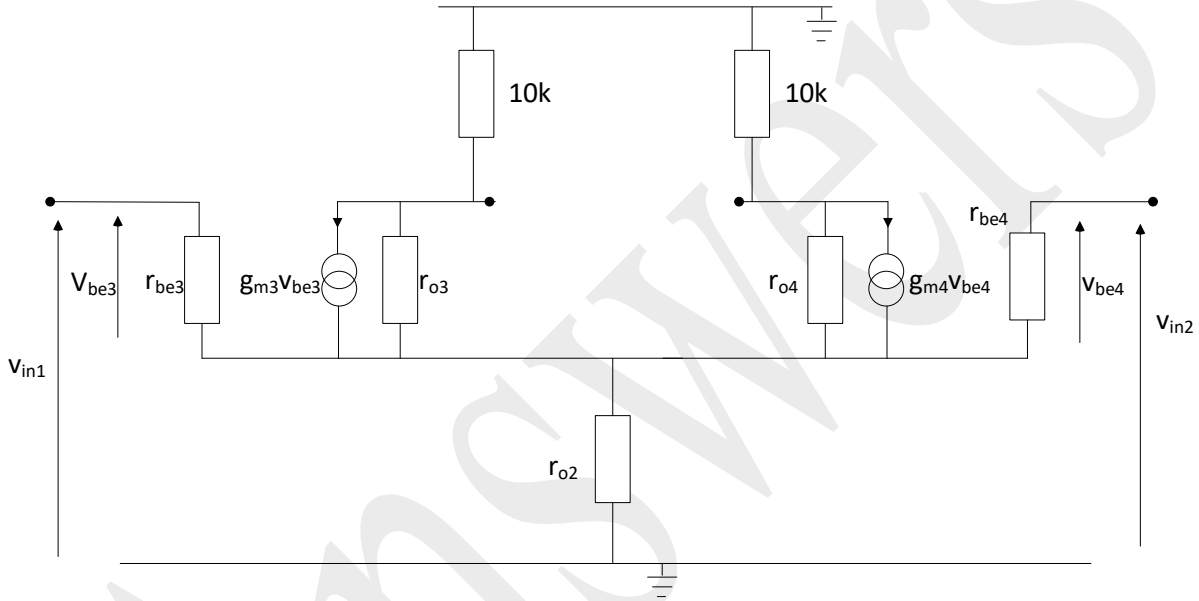
Answers

c) Draw a suitable small signal equivalent circuit of Figure 3.1 and calculate the single-ended differential-mode gain, the common-mode gain, and the common mode rejection ratio. Assume all the transistors are identical and state any assumptions you make. Assume $v_d=(v_{in1}-v_{in2})$.

[7]

g_m for each of Q3 and Q4 is I_C/V_T=0.5m/25m=0.02 S
r_{o3} and r_{o4} are 150V/0.5mA=300kΩ
r_{o2} is 150V/1mA=150kΩ
We do not need to calculate the values of r_{be}.

[2]



[2]

Under pure differential excitation the voltage at the emitters of Q3 and Q4 is a small signal ground, and the input voltages are $v_{in1}=v_d/2$ and $v_{in}2=-V_d/2$

Hence the output voltage is $v_{out}=-g_m*(r_{o4}//10k)*(-V_d/2)$ and so the gain is $-g_m*(r_{o4}//10k)/2$=-96.8 (3 s.f.)

*The common mode gain is given when $v_{in1}=v_{in2}=v_{cm}$, and is $--\alpha*10k/(2*r_{o2})$= -(100/101)*10k/2/150k=-0.033*

Thus the CMRR is the magnitude of A_d/A_{cm}= 96.8/0.033=2930 (4 s.f.))

[3]

Answers

d) What is the common-mode voltage range of the circuit in Figure 3.1 and why is it inadvisable to operate the circuit outside of these limits? What happens if these limits are exceeded?

[4]

The common-mode voltage range is the range of common-mode input voltages where the output voltage stays (almost) constant, i.e. all transistors need to be in the active mode. As vin1 and vin2, fall, Q2 will eventually saturate. This occurs when the collector is Q2 is at 0.7 V, and hence when the inputs are at 1.4 V. As the common-mode input voltage rises, Q3 and Q4 may enter saturation. This occurs when the bases of Q3 and Q4 reach 5V. hence the common mode voltage range is:
$1.4<V_{cm}<5V$

[3]

If this range is exceeded the common mode gain is no longer almost zero (the output goes to 10 V if V_{cm} falls below 1.4 V, and it follows V_{cm} with a 0.7 V lag if V_{cm} is exceeded.

[1]

e) The output of Figure 3.1 is to be capacitively coupled to the input terminal of the circuit of Figure 3.2. Firstly, calculate the voltage gain and input resistance of the circuit in Figure 3.2 in isolation. You may ignore the output resistance of the transistor in the calculation and assume that the 215 kΩ resistor is open circuit in the small signal analysis.

[6]

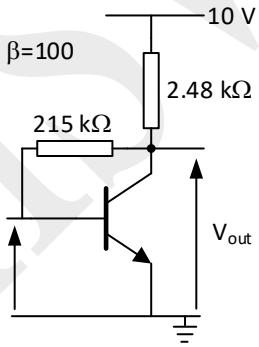


Figure 3.2

We first need to calculate the bias point of the circuit.

$I_B=(V_{out}-0.7)/215k$
$V_{out}=10-(\beta+1)I_B*2.48k$

Solving these gives $V_{out}=5.00$ V (to 3 s.f.)
The collector current is therefore $(5/2.48)*\beta/(\beta+1)=2.00$ mA (3.s.f)

[3]

This means that $g_m=2mA/25mV=0.08$ S and that $r_{be}=\beta/g_m=100/0.08=1.25$ kΩ.

Answers

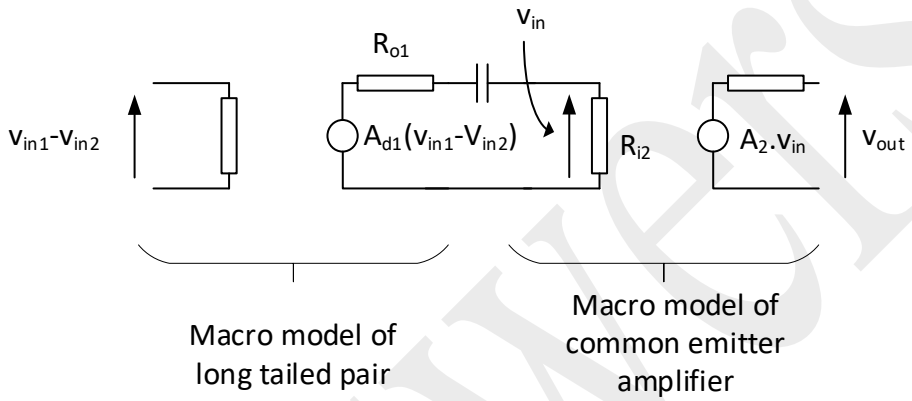
The student should not need to draw a small signal model and can simply write down that the voltage gain is $-g_m*2.48k=-198.4$ and the input resistance is simply r_{be}, or 1.25 kΩ.

[3]

f) By drawing a macro-model, calculate the overall mid-band gain of the system when Figures 3.1 and 3.2 are capacitively coupled with a 100 nF capacitor. What is the lower corner frequency of the mid-band?

[7]

We can now draw a macro model as follows:



[3]

R_{o1} is the output resistance of the differential amplifier, which is just r_{04}//10k=9.68kΩ
The mid-band gain is therefore:

$A_{d1}*(R_{i2}/(R_{i2}+R_{o1}))*A_2=-96.8*(1.25/(1.25+9.68))*-198.4=2196$ (4 s.f.)

[2]

And the corner frequency is:
$1/(2*pi*(1.25k+9.68k)*100n)=145.6$ Hz

[2]

Answers